## 4) Step-by-step: build a starter multi-room in 90 minutes

### A) AirPlay 2 "two-room" setup (iPhone + two speakers)

1. **Connect speakers to Wi-Fi** per the maker's app (HomePod, Sonos/AirPlay-capable, Denon Home, etc.).
2. On iPhone, open **Control Center → AirPlay** and tick **multiple speakers/TVs** to start in-sync playback. Apple's support guide shows the exact UI.
3. **Label rooms** in the Home app so Siri commands like "Play jazz in the Kitchen and Balcony" are natural.
4. **Stereo pair (optional):** For two HomePods in one room, Apple's guide shows how to create a left/right pair.

### B) Google Home "speaker group"

1. Install Google Home → Settings → Devices, groups & rooms → Speaker groups → choose devices → Save.
2. Cast from any supported app or Chrome to that group name; you can also transfer media to a speaker group mid-playback.

### C) Alexa Multi-Room Music

1. In the Alexa app, create a group (e.g., "Downstairs"). Amazon also creates an Everywhere group automatically.
2. Say, "Alexa, play Lata Mangeshkar everywhere," or target a specific group.

### D) Sonos "tap-to-group" plus TV sound

1. In the Sonos app, tap the group icon for a room to add others; or press-and-hold Play/Pause on any Sonos device to join what's currently playing.
2. If you use a Sonos soundbar, read Sonos' lip-sync tips (disable extra processing in the TV/source).

### E) HEOS / Denon-Marantz: send TV audio around the house

1. In the HEOS app Rooms tab, group the AVR/soundbar with other HEOS rooms.
2. Settings → My Devices → [your soundbar/AVR] → TV Sound Grouping → Enable. Tweak the sync/delay slider if you hear echo in other rooms.

## 5) Going deeper: other ways to wire a whole home

- **AVR "Zones" (wired legacy)** – many AVRs can play different sources to Zone 2/Zone 3 on speaker wire while the main room uses TV audio. Yamaha's manual explains two-zone playback. Great for terraces or a study already wired.
- **DTS Play-Fi** – a cross-brand app/platform for whole-home music with sub-millisecond sync and hi-res support; ideal if you mix brands that share Play-Fi.
- **Roon** – run a Roon Core (PC/Nucleus) and stream to Roon-Ready endpoints in perfect sync; grouping is built-in.
- **DLNA/UPnP (DIY/NAS)** – media server + multiple renderers controlled by a "control point." It works, but discovery and grouping vary.

## 6) Bluetooth LE Audio & Auracast

A new wrinkle is Auracast – part of Bluetooth LE Audio – which lets a TV or phone broadcast audio to many receivers (earbuds, hearing aids, speakers) at once. The Bluetooth SIG outlines how Auracast enables one-to-many, low-power broadcasting and shared listening.

TV makers have begun enabling Auracast on 2025 models (e.g., new LG sets), letting multiple headphones join without special hubs – handy for late-night viewing without waking up the house.

When you need perfectly synced speaker audio across rooms, Wi-Fi-based systems (AirPlay 2/Chromecast/Sonos/HEOS/MusicCast/Play-Fi) still win. For personal audio and accessibility, Auracast is a great add-on.

## 7) Volume, quality, and sync

- **Loudness/volume leveling:** Spotify provides volume normalization so tracks play at a similar loudness; it's device-specific (web/3rd-party devices may not normalize). If a room sounds "hotter," check per-app normalization and per-speaker volume.
- **AirPlay sample rates:** Apple documents Lossless tiers up to 24/48 (and Hi-Res Lossless up to 24/192 on wired gear). AirPlay-to-speaker streams often run at 44.1/48 kHz depending on device – don't chase numbers; chase stability.
- **TV audio in multiple rooms:** Grouped TV audio can introduce delay/echo in remote rooms. HEOS exposes delay controls for TV Sound Grouping; Sonos advises reducing extra TV/source processing to improve lip-sync.
- **Mesh tuning:** If a room drifts out of sync, check that each speaker is sticking to a single node (roaming can cause hiccups). Sonos and user threads mention re-assigning or wiring one unit to stabilize groups.